Geostationary Hyperspectral Infrared Sounder Channel Selection for Capturing Fast-Changing Atmospheric Information

Di Di, Jun Li, Wei Han, and Ruoying Yin

***Abstract*— Various methodologies have been developed for selecting a subset of channels from a hyperspectral infrared (IR) sounder for assimilation. The information entropy iterative method was considered optimal for channel selection. However, this method only considers the decrease in uncertainty in the atmospheric state caused by measurements at a single time, without considering the dynamic effect of measurements over a period of time; therefore, it might not be optimal for hyperspectral IR sounders onboard geosynchronous satellites that mainly aim to observe rapidly changing weather events. An alternative channel selection method is developed by adding an *M* index, which reflects the Jacobian variance over time; the adjusted algorithm is ideal for the Geosynchronous Interferometric Infrared Sounder (GIIRS), which is the first high-spectral-resolution advanced IR sounder onboard a geostationary weather satellite. Comparisons between the conventional algorithm (information entropy iterative method) and the adjusted algorithm show that the channels selected from GIIRS by the adjusted algorithm will have larger brightness temperature diurnal variations and better information content than the conventional algorithm, based on the same background error covariance matrix, the observational error covariance matrix, and the channel blacklist. The adjusted algorithm is able to select the channels for monitoring atmospheric temporal variation while retaining the information content from the conventional method. The 1-D variational (1Dvar) retrieval experiment also verifies the superiority of this adjusted algorithm; it indicates that using the channel selected by the adjusted algorithm could enhance the water vapor profile retrieval accuracy, especially for the lower and middle troposphere atmosphere.**

***Index Terms*— Channel selection, geostationary satellite, hyperspectral infrared (IR) sounder.**

I. Introduction

With the advancement of observing systems and data assimilation techniques, the positive impact of satellite measurements on numerical weather prediction (NWP) has been realized and demonstrated [1]. Among various types of measurements, the contributions from hyperspectral infrared (IR) sounders [2] are particularly outstanding. This is attributed to the fact that hyperspectral measurements contain abundant information for determining the vertical structure of atmospheric temperature and humidity at a higher accuracy. In general, thousands of channels are designed for and implemented on a given hyperspectral IR sounder; however, large data volumes and high computation loads in data processing limit the use of all channels in near-real-time (NRT) operational applications. In addition, it is not necessary to use all measurements given the high correlation and redundancy among channels.

Several data volume reduction approaches have been proposed, which attempts to identify subsets of observation data or channels that convey the essential information on the target atmospheric state parameters. Li and Wei [3] overviewed the channel selection approaches; currently, three types of data-reduction approaches for the hyperspectral sounder have been presented: selection of channels or portions of the spectrum [4]–[8] (aliased as R-method), truncation of the interferogram [9] (aliased as I-method), and truncation of the principal component (PC) scores [10], [11]. The R-method preserves the largest spectral resolution, but it destroys the spectral coverage and applies no filter to the noise. The I-method retains the full spectral coverage and reduces noise at the expense of degrading the spectral resolution. Principal component analysis (PCA) retains most of the atmospheric signal and noise is greatly suppressed. While PCA is useful for data reduction and assimilation [11], it is not considered in our channel selection comparisons since it does not use selected channels.

Our research is focused on developing an alternative suitable channel selection method for Geosynchronous Interferometric Infrared Sounder (GIIRS) onboard the FengYun (FY)-4A satellite [12], although traditional methods such as the method in [11] are still useful for GIIRS channel selection [13], [14]. FY4A is the first satellite among the second generation of the Chinese geostationary weather satellite series; the remaining satellites of this series are planned to be launched from 2021 to 2025 and beyond which also carry the hyperspectral IR sounder [12]. GIIRS is a Fourier transform spectrometer based on a classic Michelson instrument and

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measures the emitted radiance ranging from 700 to 1130 cm^{-1} (long-wave IR band) and 1650 to 2250 cm^{-1} (middle-wave IR band) at an unapodized spectral resolution of 0.625 cm^{-1}. The long-wave IR band is mainly designed to obtain the atmospheric temperature and ozone sounding, whereas the middle-wave band is used primarily for atmospheric humidity sounding, and two bands have 689 and 961 channels, respectively. GIIRS is mainly designed to observe China and adjacent regions (15°–55° N, 70°–140° E) with a spatial resolution of 16 km at nadir, and it takes 2 h to finish entire observations for the whole target region with current observation mode (after April 2019). However, higher temporal resolution (15–30 min) observations can also achieved for high-impact weather events by reducing the observation coverage. Following the conclusions drawn in [9] and [15] that the spectral resolution of GIIRS (0.625 cm^{-1}) is mandatory for temperature retrieval, the R-method receives priority. Rabier *et al.* [16] reviewed various methodologies (R-method) developed for a hyperspectral IR sounder and concluded that the information entropy iterative method from [17] is the most optimal. The main idea of this method is to select one measurement (or channel), which leads to the largest information entropy reduction (ER) on prior (or background) information in each iteration and to achieve a subset with the minimum number of channels that meet a target retrieval accuracy. The advantage of this method is that the priori information is updated accordingly before the successive channel selections in each loop, which means that this method could reduce the redundancy between channels. However, this information entropy iterative method is more appropriate for a hyperspectral IR sounder, which is operating in polar orbit than in geosynchronous orbit. An alternative channel selection method based on Jacobian variations within a time period is proposed, and this method has the potential for dynamic channel selection (e.g., based on background information for Jacobian calculations in the M index method), which could benefit the applications of geostationary hyperspectral IR sounder data, such as sounding retrieval and data assimilation.

In contrast, GIIRS, the first hyperspectral IR sounder onboard a geostationary satellite to offer both high-temporal and high-spectral resolution observations, provides continuous monitoring of the fast-changing mesoscale environment relevant to severe weather nowcasting and forecasting [18], [19]. In addition, it can improve local severe storm (LSS) short-range forecasts via regional NWP models with data assimilation [2]. Therefore, GIIRS measurements with higher sensitivity to the variability of the atmospheric state parameters are preferred; however, this is not fully considered in the information entropy iterative method.

The focus of our study is on improving the common channel selection method and adapting it to the geosynchronous hyperspectral IR sounder for application in nowcasting and NWP data assimilation. Therefore, the selected channels will have both nearly optimal information and also capture the rapid changes in atmospheric temperature and water vapor. The conventional channel selection algorithm for a hyperspectral IR sounder and the adjusted algorithm are described in Section II. The preprocessing for both algorithms is presented in Section III. The comparisons between the two algorithms are discussed in Section IV. Summary and discussions are provided in Section V.

II. Channel Selection Algorithms

The new channel selection method is based on the information entropy iterative method with adjustments and improvements specifically for a geosynchronous hyperspectral IR sounder providing observations at high temporal resolution. Hereafter, the information entropy iterative method is referred to as the conventional algorithm and the new method is referred to as the adjusted algorithm.

A. Conventional Algorithm

The theoretical background of the conventional algorithm includes an inverse and information theory. Based on the linear optimal-estimation inverse theory, the posterior or analysis error covariance matrix A of retrieval state vector x is given by

$$A = \left(H^T R^{-1} H + B^{-1}\right)^{-1} \tag{1}$$

where B is the priori error covariance matrix of the background state vector x_b and it represents the previous knowledge of the atmospheric state before any satellite measurements are applied; R is the observation error covariance matrix that consists of measurement errors and radiative transfer model (RTM) simulation errors; and H is the Jacobian matrix containing partial derivatives of the channel measurements with respect to the atmospheric state vector. Provided that the R matrix is diagonal and a channel is selected in each iteration, the analysis error covariance matrix A_i (after $i-1$ channels have been chosen) could be rapidly calculated by

$$A_i = A_{i-1}\left[I - \frac{h_i (A_{i-1} h_i)^T}{1 + (A_{i-1} h_i)^T h_i}\right] \tag{2}$$

where A_i starts with $A_0 = B$, h_i is the Jacobian matrix of channel i that is normalized by $R^{-1/2}$, and I is the identity matrix. It should be noted that using the diagonal observation error covariance matrix for hyperspectral IR sounders with apodization process is not a good assumption and more recent channel selection work evaluated the full analysis error covariance each time [20], [21], which allows considering the off-diagonal full R-matrix. For simplicity, we still use the diagonal component consistently in both the conventional and adjusted methods for comparisons.

The appropriate quantity to describe the information gained by one measurement was defined by Shannon and Weaver [22] as the "information entropy." Information entropy can be interpreted as the logarithm of the volume of state space within a particular contour of the probability density function

$$H(p) = -\int p(x) \log\left(\frac{p(x)}{M}\right) dx. \tag{3}$$

Reduction on entropy means information gain. Also, the information content of a measurement can be quantified by the change in entropy consequent on making the measurement. Information content is thus directly related entropy to a given atmospheric state described by its probability

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density function. In the case of a Gaussian probability density function, Rodgers [23] demonstrated that the entropy ascribed to the covariance S of vector x can be written as $H(p) = (1/2)\log|S|$.

The conventional channel selection algorithm, based on the information content of the measurements, reflecting the reduction in the analysis error (A) over the background error (B), could be quantified by a degree of freedom for signals (DFS) or the ER

$$\mathrm{DFS} = tr\left(I - AB^{-1}\right) \tag{4}$$

$$\mathrm{ER} = -\frac{1}{2}\ln\left(AB^{-1}\right). \tag{5}$$

Both quantities give the reduction of background atmospheric status uncertainty due to the measurements as explained in [17]. Rabier *et al.* [16] showed that there are almost no channel selection differences when applying DFS or ER as the figure of merit, and ER is adopted in this study. Here, the goal is to select the temperature and water vapor sensitive channels, and the information content of the measurements on temperature and water vapor is the focus.

Collard [5] modified the information entropy iterative method proposed by Rabier *et al.* [16] by prescreening the selection of infrared atmospheric sounding interferometer (IASI) channels. There are two main stages for this selection algorithm: 1) removal of channels based on *ad hoc* criteria in a prescreening stage and 2) individual channel selection using the information entropy iterative method with the blacklisted channels excluded. Collard [5] listed some factors to be considered in prescreening, including trace species, intercomparisons (brightness temperature (BT) differences between different RTMs), and NWP monitoring statistics that are very useful for channel selection. Note that, two adjacent channels will not be chosen together to avoid a larger channel correlation since the Hamming apodization is applied in the simulation of GIIRS radiance spectra [24].

B. Adjusted Algorithm

The adjusted algorithm aims to select the GIIRS channels that are sensitive to the variability of the atmospheric state. First, a new index (named the M index) is introduced in the adjusted algorithm, which is used to objectively evaluate the Jacobian variability over time. It is defined in the following formula:

$$M = \min(l_1, l_2)\sqrt{\frac{\sum_{i=1}^{\min(l_1,l_2)}\left(J_{2,i} - J_{1,i}\right)^2}{\sum_{i=1}^{\min(l_1,l_2)} J_{1,i}^2}} \tag{6}$$

where $J_{1,i}$ and $J_{2,i}$ represent the Jacobian matrices at pressure level i at two consecutive times, respectively, and l_1 and l_2 represent the effective pressure levels (the pressure levels above surface pressure) for $J_{1,i}$ and $J_{2,i}$, respectively. The concept of the M index is adapted from [25], and it was originally used to evaluate the differences or the uncertainty in Jacobian matrices calculated from various RTMs.

The quantitative M indices for each GIIRS channel are calculated using the radiative transfer for TOVS (RTTOV) model [26]–[28] and the ERA-5 reanalysis [29] created at the European Center for Medium-Range Weather Forecasts (ECMWF) with a temporal resolution of 1 h and a spatial resolution of 0.25°. The GIIRS coefficients used in RTTOV are generated using a weighted least square method with local training profiles [24]. Specifically, ERA-5 reanalysis data for one day (April 1, 2018) over the GIIRS observing region (15°–55° N, 70°–140° E) are used as input to simulate Jacobian matrices with the GIIRS RTTOV_K model. Jacobians at two consecutive hours for each location are then used to obtain M indices by (6). M indices are averaged temporally for each location and spatially averaged for the whole region. Finally, spatially and temporally averaged M indices are calculated for each GIIRS channel. The M indices, as well as the pressures of Jacobian peaks for the most temperature or water vapor sensitive channels, are shown in Fig. 1. Note that the Jacobian peak heights/pressures (red dots in Fig. 1) are calculated with the U.S. standard atmospheric profile and GIIRS RTTOV_K model for illustration. By the way, for further analyses, the selection range of temperature channels that refer to [5] is marked. It can be seen that the M index is comparatively larger for channels that are sensitive to the low troposphere temperature and smaller for channels that are sensitive to the upper atmosphere temperature, while for a water vapor channel, the M index does not vary with the Jacobian peak height. The temperature Jacobian function of a temperature channel reflects the atmospheric radiation sensitivity, which is highly related to atmospheric temperature change at each pressure level. Usually, temperature has larger variation in the lower/middle troposphere than in the upper troposphere; therefore, the M indices for lower/middle troposphere channels are larger than those for the upper tropospheric channels. In contrast, the water vapor channels show the similar M index regardless of the peak height of the channel weighting function, and there are two possible reasons: 1) the water vapor is mainly concentrated in the middle/lower troposphere, the water vapor Jacobian function of a water vapor absorption channel reflects the sensitivity of radiance to the water vapor

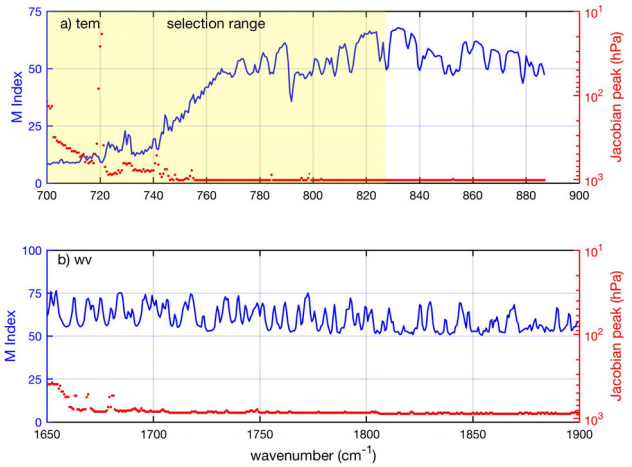


Fig. 1. M index (blue lines) and pressure of Jacobian peak (red dots) for main (a) temperature channels and (b) water vapor channels. M indices are calculated with ERA-5 reanalysis data. The Jacobians are calculated with the U.S. standard atmospheric profile.

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change at each pressure level, and moisture variation happens in the whole troposphere and 2) it is difficult to separate the contributions of the surface radiation and the near-surface atmospheric radiation, respectively, in the radiance observation of a water vapor absorption channel unless there is a large contrast between surface skin temperature and surface air temperature. Therefore, the boundary layer water vapor information is limited in the water vapor channel radiance under the most atmospheric conditions [30].

With the conventional algorithm, the channels are compared simultaneously and selection is on a single-channel basis with the largest information ER according to priori or background information in each iteration. In the conventional algorithm, only one channel with the largest ER through the information entropy iterative method is selected in each iteration. The adjusted algorithm combines the conventional algorithm with the channel M index. It aims to select the channels that maximize the information gain on the background atmospheric state and also contains the greatest sensitivity to atmospheric state variability. In the adjusted algorithm, first 20 channels with the largest ER are preliminarily selected in each iteration through the information entropy iterative method; among these 20 channels, the channel with the largest normalized M index will then be the final selection in this iteration. Once a channel is selected, this iteration ends. The entire selection procedure ends when the target number of channels is reached. It should be mentioned that it is only an alternative combination of the conventional algorithm and M index. More channels with the largest ER could be compared in each iteration for selection, depending on the priority for information content or atmospheric sensitivity of the selected channels. Comparing 20 channels in each iteration enables the selection of more channels sensitive to atmospheric variance without losing much information content.

Note that, in addition to the M index, the adjusted algorithm also makes further improvements during the prescreening stage. For example, additional factors, such as the Jacobian nonlinearity and obvious Jacobian double/multiple peaks, are included in the channel selection since channels with higher Jacobian nonlinearity or multiple peaks will influence its further application in NWP data assimilation and sounding retrieval. The definition of Jacobian nonlinearity and the process of blacklisting channels will be further explained in Section III-A.

III. Selection of GIIRS Channels

Both the conventional algorithm and the adjusted algorithm require an estimate of the background error covariance matrix for temperature and other absorption gas species (e.g., water vapor), as well as an estimate of the observational error covariance matrix. Moreover, the channel blacklist for both algorithms must be the same; an explicit description is provided in this section.

A. Channel Blacklist

The main purpose of the prescreening stage is to avoid channels with larger forward model uncertainty, including channels that have: 1) trace gas absorption whose variability is not well depicted in the RTM; 2) larger simulation errors or uncertainties among various RTMs; and 3) higher Jacobian nonlinearity or multiple Jacobian peaks.

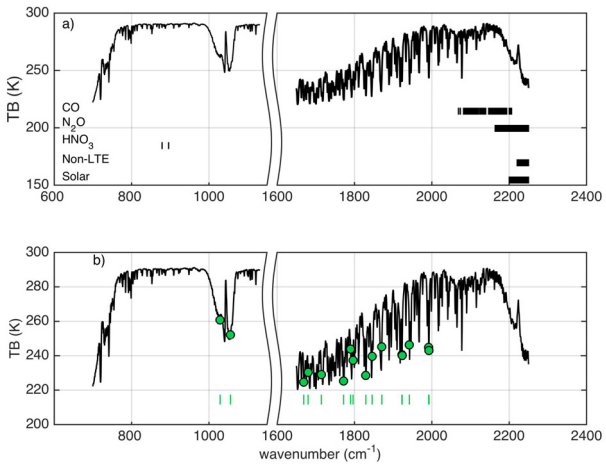


Fig. 2. Blacklisted channels that have (a) trace gas absorption such as CO, N_2O, HNO_3, or impacts from solar irradiance and non-LTE (marked with solid line) and (b) larger model BT simulation errors (marked with green circle).

In step 1, channels with strong trace gas absorption are removed. Following Collard [5], ten trace gas species have been tested, including CH_4, CO, N_2O, CCL_4, CFC-11, CFC-12, CFC-14, HNO_3, NO_2, OCS, NO, and SO_2. If the impact of any trace gas on a given channel's BT is larger than 1k for any of six typical atmospheric profiles as determined by the line-by-line radiative transfer model (LBLRTM) [31], this channel will be blacklisted (not used). If the impact of the trace gases on any channel is lower than 1k, it will be added into the observational error covariance matrix. In this step, channels that might be influenced by solar irradiance or un-modeled non-local thermodynamic equilibrium (LTE) [32] will also be excluded. The results are shown in Fig. 2(a).

In step 2, channels that have larger RTTOV model simulation uncertainty are excluded. The RTTOV model is assessed using the LBLRTM model with 200 independent test profiles (for more details, refer to [24]). In addition, Di *et al.* [24] generated two sets of GIIRS coefficients with different training profiles. The BT simulation differences from the two sets of GIIRS coefficients are also used as a criterion here. Those channels with larger BT differences are also blacklisted. The channels removed in this step are marked in Fig. 2(b).

In step 3, to account for the Jacobian nonlinearity, the Jacobian is calculated with different perturbations in RTTOV. The Jacobian differences between 10% and 30% perturbations in water vapor are used as an indicator of nonlinearity for water vapor; in addition, 0.5k and 1.5k perturbations in temperature are evaluated. Here, six typical atmospheric profiles are used in the RTTOV model to check the Jacobian. Jacobian differences can be quantified by the M value described in [25], where a goodness of fit measurement, M, was defined by (6). Smaller M values indicate lower nonlinearity in the Jacobians. Refer to [25], the threshold of 25k indicates a quite inconsistent

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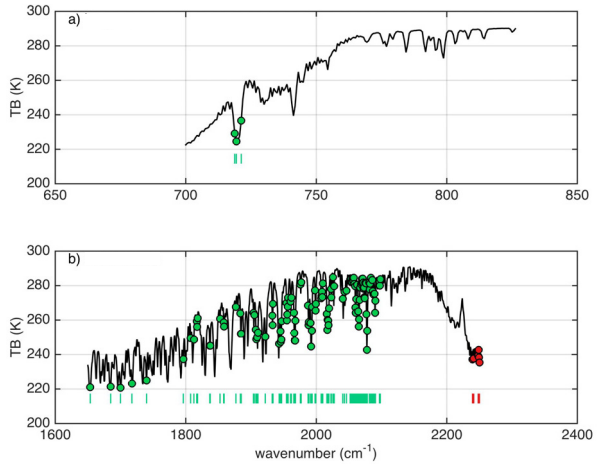


Fig. 3. Channel blacklist for those channels that have higher nonlinearity or multiple peaks in Jacobians. Green represents the channels blacklisted due to multiple peaks and red represents the channels blacklisted due to higher nonlinearity. (a) Result of long-wave band. (b) Result of middle-wave band.

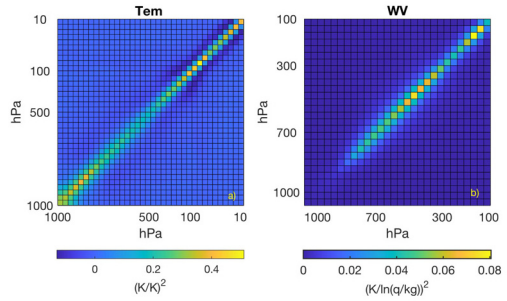


Fig. 4. Background error covariance matrix for (a) temperature and (b) water vapor.

Jacobian fit, which reflects larger Jacobian nonlinearity here. Since a relatively larger threshold is used, only six shortwave temperature channels are excluded [see red dots in Fig. 3(b)]. In this step, channel Jacobians that have obvious double or multiple peaks are also excluded. It is a relatively straightforward process: temperature and water vapor mixing ratio (lnq) Jacobians are calculated with the RTTOV_K model and six typical atmospheric profiles, and channel Jacobians are then checked with respect to their patterns. The channel blacklist in this step is shown in Fig. 3.

B. Background Error Covariance Matrix

The temperature and water vapor background error covariance matrix is taken from the 1DVar package created at the European Organization for the Exploitation of Meteorological Satellites NWP Satellite Application Facility (EUMETSAT NWPSAF). Both background error covariance matrices for temperature and water vapor are plotted in Fig. 4.

A diagonal observational error covariance matrix is used in our study. Its diagonal errors include the noise equivalent delta temperature (NEdT) of GIIRS, the forward model simulation errors, and the effects of trace gases. The NEdT is calculated from the averaged noise equivalent delta radiance (NEdR) within GIIRS L1B data for one month (August 2017).

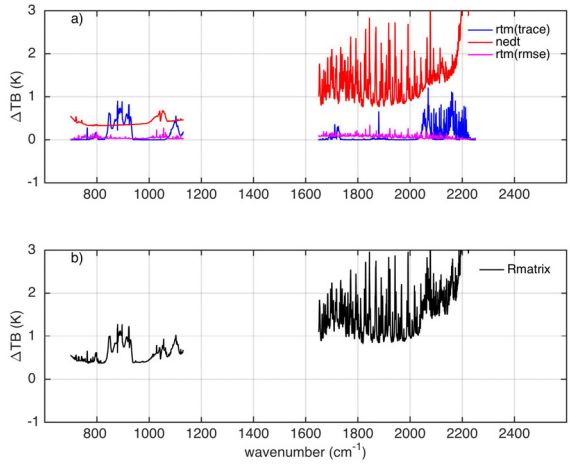


Fig. 5. (a) Elements of the observational error covariance matrix include NEdT, forward RMSE, and the effect of trace gases. (b) Square root of the diagonal observational error covariance matrix.

The forward model simulation error of RTTOV is represented by the root-mean-square error (RMSE) calculated from 200 independent test profiles using the LBLRTM model as a reference [24]. The specific procedure for calculating the effect of trace gases is given in Section III-A. Fig. 5(a) shows three types of measurement or forward errors, and Fig. 5(b) shows the sum of the three errors which is the square root of the diagonal elements of the observational error covariance matrix.

IV. Results and Analysis

As suggested by Collard [5], temperature information should mainly come from CO_2 absorption channels rather than from H_2O and O_3 channels. The temperature channel selection is restricted to 700–827 cm^{-1} with the blacklisted channels removed, while the water vapor channel selection is mainly limited to 1650–2100 cm^{-1}.

Based on the same background and observational error covariance matrices and using the same channel blacklist, channel selection is performed with both the conventional algorithm and the adjusted algorithm, respectively. The ER is taken as the figure of merit. The selection strategy is to find the minimum number of channels containing sufficient target parameter retrieval accuracy. In addition, since the Hamming apodization is applied in the RTTOV simulation; if one channel is chosen, its two adjacent channels cannot be chosen.

A. Results of Single Profile

Comparisons between two algorithms are initially based on a single profile (e.g., the U.S. standard atmospheric profile), which helps to analyze the differences between the two algorithms in detail. Two sets of temperature and water vapor-sensitive channels are chosen by the conventional algorithm and the adjusted algorithm, respectively. The number of selected temperature channels is fixed at 65 for both algorithms. Since the temperature channel selection is restricted to 700–827 cm^{-1} with three blacklisted channels

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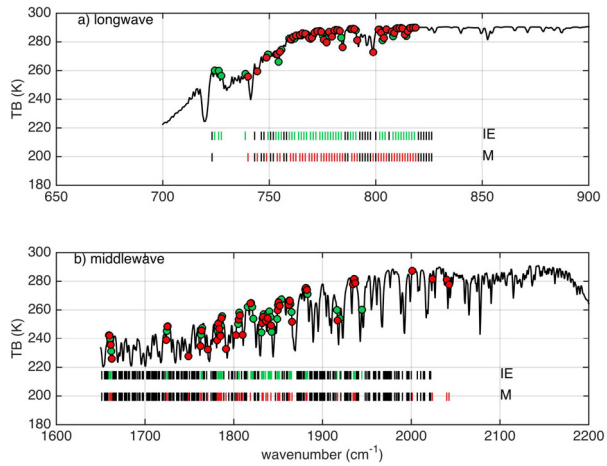


Fig. 6. Spectral locations for distinct (a) temperature channels and (b) water vapor selected by the two algorithms. The green dots and the red dots represent the channels selected by the conventional algorithm and the adjusted algorithm, respectively.

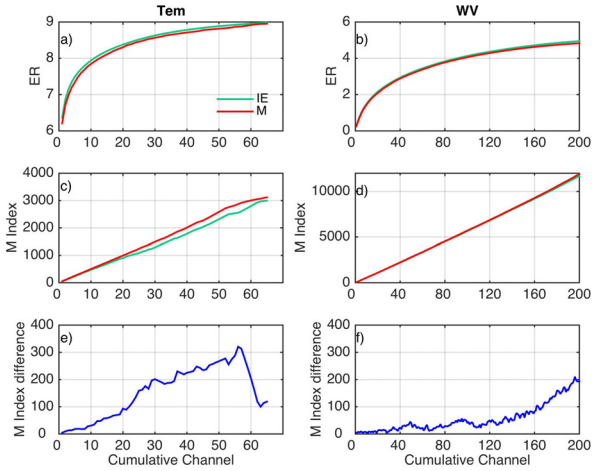


Fig. 7. (a) and (b) Accumulated ER and (c) and (d) M indices for two sets of temperature channels and water vapor-sensitive channels. The green solid lines and the red solid lines represent the results from the conventional algorithm and the adjusted algorithm, respectively. The M index differences for (e) temperature and (f) water vapor channels are separately plotted.

removed [see Fig. 3(a)], there are only 200 temperature channels left for selection. Also, considering the restrictions with Hamming apodization of GIIRS spectra, two adjacent channels will not be chosen together, and 65 (about one-third of candidate temperature channels) is the maximum number of temperature channels that can be selected. With 323 channels blacklisted in Fig. 3(b), there are only 620 water vapor channels kept. For the same reason due to Hamming apodization, the number of humidity channels is set at 200. The spectral locations and Jacobian patterns of the two sets of temperature or water vapor sensitive channels both show sensitivity for various atmospheric levels (not shown). The green dots/line and the red dots/line in Fig. 6 represent the distinct channels selected by the conventional algorithm and the adjusted algorithm, respectively. Also, black short solid lines mark the spectral position of the common channels. While there are obvious temperature channel differences (42 distinct temperature channels), there are comparatively fewer water vapor channel differences (41 distinct water vapor channels) between the two algorithms. To be specific, 42 distinct temperature channels mean that the two algorithms get 42 different channels when they both select 65 channels in this case. The major differences between the two algorithms for temperature channel selection are in the lower troposphere, while the differences for water vapor channel selection are in the middle troposphere. The similarity in channel selection for both temperature and water vapor suggests that the conventional algorithm also has some skill in choosing the channels sensitive to atmospheric variability.

The major contrast between the algorithms is illustrated by two indices: ER index and M index. Accumulated ER and M indices of consecutive channels chosen by the two algorithms have been calculated. The accumulated ER index represents the information content contained in the selected channels. The accumulated M index indicates the sensitivity of the channels with respect to atmospheric variability. The indices for different temperature and water vapor-sensitive channels selected by the two algorithms are shown in Fig. 7. In Fig. 7(a) and (b), for the accumulated ER, it can be seen that there are almost no information content differences between the two sets of temperature or water vapor channels. This result can partly be attributed to the fact that the modification to the conventional algorithm is comparatively modest. It also implies that the information content differences among channels are relatively small. In Fig. 7(c) and (d), for the M index, however, the differences between both algorithms are discernible, especially for the temperature channels. Fig. 7(e) reveals that when sufficient channels are selected by the conventional algorithm, the sensitivity differences to atmospheric temperature variability between the two algorithms might decrease. The M index differences between the two sets of water vapor channels are not obvious since the M index between each water vapor channels is small [shown in Fig. 1(b)]. However, the algorithm still has selected the water vapor channel with a comparatively higher M index.

In addition, M index differences are shown as BT measurement differences. The BT diurnal variation differences between the two sets of channels selected by the two algorithms are further compared. Here, ERA-5 reanalysis data for one day (April 1, 2018) with 1-h temporal resolution over the GIIRS observing region (15°–55° N, 70°–140° E) are used to simulate the BTs. Then, spatially averaged BTs over the whole region for each channel are obtained. Similarly, comparisons are also made between distinct channels (in this study, 42 distinct temperature channels and 41 water vapor channels). Here, the BT anomaly instead of BT itself is plotted to intuitively show the BT diurnal variation. BT anomaly is calculated by subtracting the mean (averaged BT from 24 h) from the BT at each hour. Diurnal variations in BT anomalies of temperature and water vapor distinct channels, with area- and channel-averaged, are plotted in Fig. 8. The green solid lines and the red solid lines represent the results from the conventional algorithm and the adjusted algorithm,

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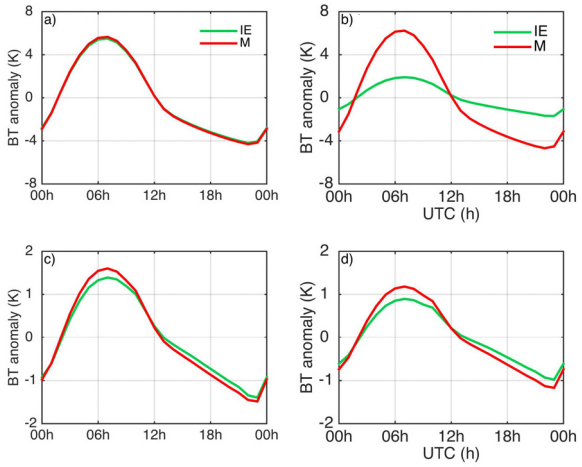


Fig. 8. Diurnal variations in BT anomaly for (a) temperature and (c) water vapor distinct channels selected by two algorithms using the U.S. standard atmospheric profile. The same result for (b) temperature and (d) water vapor distinct channels selected by the two algorithms using the typical atmospheric profile of a middle-latitude winter. The BT simulations are averaged for the distinct temperature and distinct water vapor channels over the GIIRS observation region. The green solid lines and the red solid lines represent the results from the conventional algorithm and the adjusted algorithm, respectively.

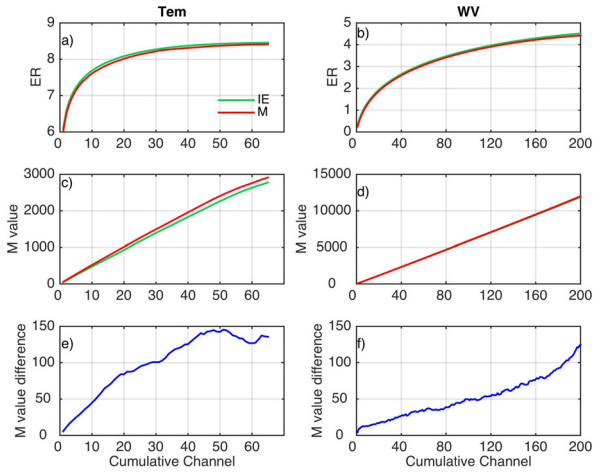


Fig. 9. Mean (a) and (b) accumulated ER and (c) and (d) *M* indices for two sets of temperature channels and water-sensitive channels based on 67 profiles, respectively. The green solid lines and the red solid lines represent the results from the conventional algorithm and the adjusted algorithm, respectively. The *M* index differences for (e) temperature and (f) water vapor channels are separately plotted.

respectively. The amplitude of the water vapor channels from the adjusted algorithm is larger—approximately an amplitude increasement of 0.5k—whereas the diurnal variation amplitude of the temperature channel is almost the same. As analyzed above, since enough temperature channels (almost the upper limit of the number of selectable temperature channels) are selected by both algorithms, the superiority of the adjusted algorithm is not reflected here. When the number of selected temperature channels decreases, the amplitude of the temperature channels from the adjusted algorithm is larger (not shown). Even when selecting the same number of channels using a colder atmospheric profile (e.g., typical atmospheric profile from a middle-latitude winter), the amplitude of the distinct temperature channel selected by the adjusted algorithm is obviously larger than that of the conventional algorithm (approximately an amplitude increasement of 4.5 K). In conclusion, the improvement from the adjusted algorithm is more obvious when fewer target channels are selected or the background atmospheric condition is comparetively cold. Xue *et al.* [33] analyzed the water vapor diurnal variation using both the reanalysis and the high-temporal-resolution geostationary water vapor radiance measurements, and both sources of data reveal the water vapor diurnal cycle. Therefore, selected channels that reflect diurnal variation would benefit applications, such as situation awareness using either retrievals or radiance measurements from the selected channels.

B. Selection Results Using a Set of Profiles

The robustness of the adjusted algorithm must be ascertained. Thus, a set of 67 profiles are used to compare the differences between the adjusted algorithm and the conventional algorithm. The profiles were sampled from the SeeBor Version 5.0 database created at the Cooperative Institute for Meteorological Satellite Studies (CIMSS), University of Wisconsin–Madison [34]. The profile sampling procedure was developed by Di *et al.* [24]. The profile set is representative of various atmospheric states within the GIIRS coverage. First, each profile is used to select the corresponding 65 temperature and 200 water vapor sensitive channels with the two algorithms. The accumulated ER index and *M* index for the two algorithms based on each profile are calculated (as shown in Fig. 7) and then averaged for all the profiles. Fig. 9 shows the mean accumulated ER index and *M* index for the two algorithms based on the 67 profiles. The results are displayed separately for temperature and water vapor channels. Similar conclusions could be drawn from Fig. 7. The adjusted algorithm does not deviate from the information content in terms of ER index and *M* index for the selected channels, while it will pick up the channels that have more sensitivity to atmospheric state variability than those from the conventional algorithm.

Similarly, diurnal variations in BT anomalies between temperature and water vapor channels selected by the two algorithms based on the 67 profiles are also compared. Specifically, distinct temperature and water vapor channels could be obtained for each profile. The diurnal variations in BT anomalies for distinct temperature and water vapor channels are averaged and shown in Fig. 10(a) and (c). The green solid lines and the red solid lines represent the results from the conventional algorithm and the adjusted algorithm, respectively. Similar conclusions could be drawn from Fig. 10(a) and (c) as from Fig. 8. In addition, the diurnal variations in BT anomalies for certain profiles are also shown. As shown in Fig. 10(b) and (d), using this profile to select channels, there are amplitude increases of approximately 5k for temperature and 1k for water vapor using the adjusted algorithm over the conventional algorithm, confirming the advantages of the adjusted algorithm.

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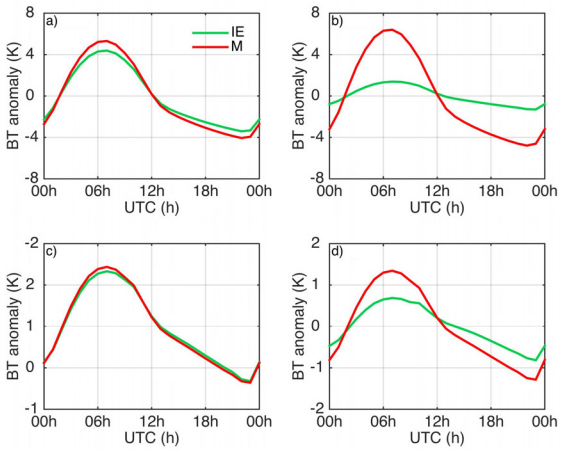


Fig. 10. Diurnal variations in BT anomalies of (a) temperature and (c) water vapor distinct channels, in which BTs are averaged in area and channels. The results are calculated from 67 profiles. The same results for (b) temperature and (d) water vapor distinct channels but based on one of the 67 profiles. The green solid lines and the red solid lines represent the results from the conventional algorithm and the adjusted algorithm, respectively.

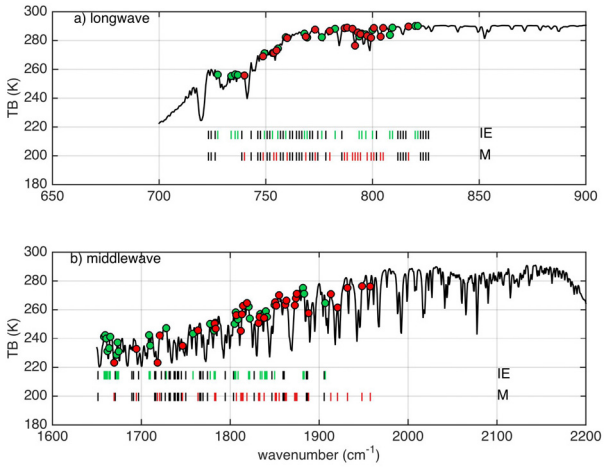


Fig. 11. Spectral locations for 50 temperature and 60 water vapor channels selected by (a) conventional algorithm and (b) adjusted algorithm, respectively.

Please note that it worth noting that the difference of diurnal variation of BT anomaly for the temperature channels between the adjusted algorithm and the conventional algorithm is partly due to some upper/middle tropospheric temperature channels that are included in the channels set selected by the conventional algorithm, which are distributed over 700–750 cm^{-1} (see Fig. 6), considering that the BT anomaly is computed by averaging the BT anomaly for all selected temperature channels. In general, the temperature diurnal variation is mainly shown in the lower troposphere because the radiation emitting from the surface, heated by the solar radiation, warms the ambient temperature in the lower troposphere. Therefore, including the upper/middle-tropospheric channels in the computation of BT diurnal variation makes the pattern of diurnal variation smooth. Thus, if the upper/middle-tropospheric temperature channels, less sensitive to the lower tropospheric temperature, for the channel set selected by the conventional algorithm are removed, the significant difference of the diurnal variation between two-channel sets is reduced (results not shown).

In general, a "constant" channel selection must be determined based on different atmospheric profiles in an off-line study and used for operational applications. The iterative selection is expensive and would be prohibitive to perform in real-time operations. Here, two sets of constant GIIRS channels are selected off-line by both algorithms based on the abovementioned 67 local profiles. The method to assemble a set of constant channels based on various atmospheric profiles follows [15]. For further examination, a 1-D variational (1Dvar) retrieval experiment has been conducted to analyze the retrieval accuracy difference between two sets of constant GIIRS channels separately selected by the two algorithms. In retrieval experiments, the ERA5 reanalysis dataset is designed as the test profiles (the number of profiles is 69864) and each test profile has added random forecast errors. The forecast errors are from the background error covariance matrix (shown in Fig. 4) and it can be calculated by

$$x_0 = x_t + \sum_i \varepsilon_i \lambda_i^{\frac{1}{2}} E_i \tag{7}$$

where x_0 is the background forecast profile, x_t is the truth profile (ERA5 reanalysis data), λ_i and E_i are the eigenvalue and eigenvector of the background error covariance matrix at pressure level i, respectively, and ε_i is the random Gaussian error with a zero mean and a unit standard deviation. In addition, instrument noise errors derived from the observation error matrix (shown in Fig. 5) are also added to simulate GIIRS observations. Note that in this experiment, the constant GIIRS channels only consist of 50 temperature channels and 60 water vapor channels (shown in Fig. 11). When choosing the selectable maximum number of channels (65 temperature channels and 200 water vapor channels), retrieval accuracies for both sets of channels are completely equivalent (not shown). The superiority of the adjusted algorithm might be fully reflected when fewer channels are used in the retrieval/DA system.

First, the background profiles are designed with the original hourly ERA5 reanalysis dataset, which means that the background forecast profiles are highly accurate. The temperature and water vapor profile retrieval results using two sets of channels are shown in Fig. 12(a) and (b), respectively. It suggests that the temperature retrieval accuracy between these algorithms is equivalent, but water vapor retrieval accuracies are enhanced, especially for middle and lower troposphere water vapor. In addition, when the background profiles are designed with the hourly averaged ERA5 reanalysis dataset, that is to say, the accuracies of the background forecast profiles are partly decreased, the water vapor profile retrieval accuracies using the channels from the adjusted algorithm are also enhanced. The retrieval improvements mainly occur in the middle and lower troposphere, which relates to the adjusted algorithm tending to pick out channels sensitive to the atmosphere in the middle and lower troposphere.

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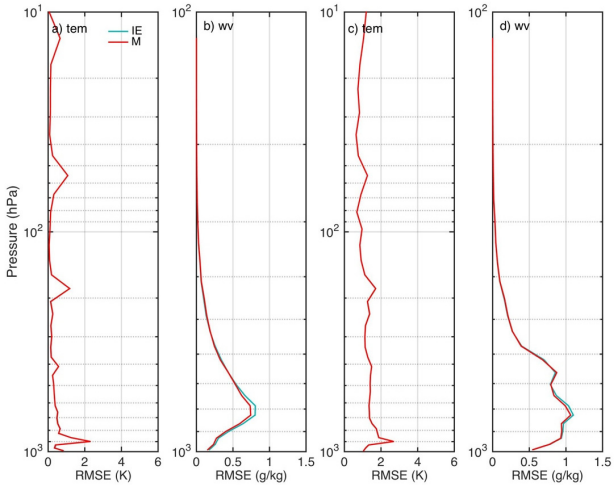


Fig. 12. (a) Temperature and (b) water vapor retrieval RMSE for two sets of constant channels selected by the two algorithms, using the original hourly ERA5 reanalysis dataset as the test profiles. Similar results for (c) temperature and (d) water vapor retrieval RMSE but using the hourly averaged ERA5 reanalysis dataset as the test profiles.

V. Conclusion and Discussion

An alternative channel selection algorithm for GIIRS, the first hyperspectral IR sounder onboard a geostationary satellite, has been developed. The new algorithm is adjusted from the commonly used information entropy iterative method (the conventional algorithm) by introducing a new index representing the Jacobian variability over a period of time. The adjusted algorithm is able to select channels that capture the rapid changes in atmospheric temperature and humidity.

In this study, we found that the following conditions hold.

1) Based on the same background error covariance matrix, the observational error covariance matrix, and the channel blacklist, the channels selected by the adjusted algorithm will have larger BT diurnal variations and much closer information content than the conventional algorithm. It means that the adjusted algorithm could improve the ability to select the best channels for monitoring atmospheric temporal variation while retaining the information content from the conventional algorithm. It might apply well in selecting geostationary hyperspectral IR channels for various applications, such as severe weather nowcasting (e.g., convection recognition).
2) 1Dvar experiments indicate that using the channel selected by the adjusted algorithm could enhance the water vapor profile retrieval accuracy, especially for the lower and middle troposphere atmosphere.

The M index preprocessing in the adjusted channel selection algorithm will be further calculated in other seasons and the adjusted channel selection algorithm will be tested in data assimilation. This new algorithm is able to select channels that are more suitable for capturing the fast-changing weather events from the current and future hyperspectral IR sounders onboard geostationary satellites, such as those onboard the Chinese FengYun-4 series and the future Meteosat Third Generation (MTG) series [35]. However, note that this channel selection method has some limitations and needs further improvement, for example, it does not consider the cloud effect, with future "all-sky" IR radiance assimilation likely, and it is important to select channels also suitable for assimilation under cloudy skies. Besides, the development of the full analysis error covariance, which allows the off-diagonals in the full R-matrix, is also the future work.

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References

[1] P. Bauer, A. Thorpe, and G. Brunet, "The quiet revolution of numerical weather prediction," *Nature*, vol. 525, no. 7567, pp. 47–55, Sep. 2015.
[2] W. P. Menzel, T. J. Schmit, P. Zhang, and J. Li, "Satellite-based atmospheric infrared sounder development and applications," *Bull. Amer. Meteorolog. Soc.*, vol. 99, no. 3, pp. 583–603, Mar. 2018.
[3] J. Li and W. Han, "A step forward toward effectively using hyperspectral IR sounding information in NWP," *Adv. Atmos. Sci.*, vol. 34, no. 11, pp. 1263–1264, Nov. 2017.
[4] N. Fourrié and J.-N. Thépaut, "Evaluation of the AIRS near-real-time channel selection for application to numerical weather prediction," *Quart. J. Roy. Meteorolog. Soc.*, vol. 129, no. 592, pp. 2425–2439, Jul. 2003.
[5] A. D. Collard, "Selection of IASI channels for use in numerical weather prediction," *Quart. J. Roy. Meteorolog. Soc.*, vol. 133, no. 629, pp. 1977–1991, Oct. 2007.
[6] A. Gambacorta and C. D. Barnet, "Methodology and information content of the NOAA NESDIS operational channel selection for the cross-track infrared sounder (CrIS)," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 6, pp. 3207–3216, Jun. 2013.
[7] Y.-C. Noh, B.-J. Sohn, Y. Kim, S. Joo, W. Bell, and R. Saunders, "A new infrared atmospheric sounding interferometer channel selection and assessment of its impact on met office NWP forecasts," *Adv. Atmos. Sci.*, vol. 34, no. 11, pp. 1265–1281, Nov. 2017.
[8] L. Jun and H. Hung-Lung, "Optimal use of high resolution infrared sounder channels in atmospheric profile retrieval," *Adv. Atmos. Sci.*, vol. 11, no. 3, pp. 271–276, Sep. 1994.
[9] G. Grieco, G. Masiello, and C. Serio, "Interferometric vs spectral IASI radiances: Effective data-reduction approaches for the satellite sounding of atmospheric thermodynamical parameters," *Remote Sens.*, vol. 2, no. 10, pp. 2323–2346, Sep. 2010.
[10] P. Antonelli *et al.*, "A principal component noise filter for high spectral resolution infrared measurements," *J. Geophys. Res., Atmos.*, vol. 109, no. D23, pp. 1–22, Dec. 2004.
[11] A. D. Collard, A. P. McNally, F. I. Hilton, S. B. Healy, and N. C. Atkinson, "The use of principal component analysis for the assimilation of high-resolution infrared sounder observations for numerical weather prediction," *Quart. J. Roy. Meteorolog. Soc.*, vol. 136, no. 653, pp. 2038–2050, Oct. 2010.
[12] J. Yang, Z. Zhang, C. Wei, F. Lu, and Q. Guo, "Introducing the new generation of Chinese geostationary weather satellites, Fengyun-4," *Bull. Amer. Meteorolog. Soc.*, vol. 98, no. 8, pp. 1637–1658, Aug. 2017.
[13] R. Yin, W. Han, Z. Gao, and G. Wang, "A study on longwave infrared channel selection based on estimates of background errors and observation errors in the detection area of FY-4A (in Chinese)," *Acta Meteor. Sinica*, vol. 77, no. 5, pp. 898–910, 2019.
[14] R. Yin, W. Han, Z. Gao, and D. Di, "The evaluation of FY4A's geostationary interferometric infrared sounder (GIIRS) long-wave temperature sounding channels using the GRAPES global 4D-Var," *Quart. J. Roy. Meteorolog. Soc.*, vol. 146, no. 728, pp. 1459–1476, Apr. 2020.
[15] Q. Liu and S. Xiao, "Effects of spectral resolution and signal-to-noise ratio of hyperspectral sensors on retrieving atmospheric parameters," *Opt. Lett.*, vol. 39, no. 1, pp. 60–63, 2014.
[16] F. Rabier, N. Fourrié, D. Chafaï, and P. Prunet, "Channel selection methods for infrared atmospheric sounding interferometer radiances," *Quart. J. Roy. Meteorolog. Soc.*, vol. 128, no. 581, pp. 1011–1027, Apr. 2002.

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[17] C. D. Rodgers, "Information content and optimisation of high spectral resolution remote measurements," *Adv. Space Res.*, vol. 21, no. 3, pp. 361–367, Jan. 1998.
[18] J. Li, J. Li, J. Otkin, T. J. Schmit, and C.-Y. Liu, "Warning information in a preconvection environment from the geostationary advanced infrared sounding system—A simulation study using the IHOP case," *J. Appl. Meteorolog. Climatol.*, vol. 50, no. 3, pp. 776–783, Mar. 2011, doi: 10.1175/2010JAMC2441.1.
[19] J. Li, C.-Y. Liu, P. Zhang, and T. J. Schmit, "Applications of full spatial resolution space-based advanced infrared soundings in the preconvection environment," *Weather Forecasting*, vol. 27, no. 2, pp. 515–524, Apr. 2012, doi: 10.1175/WAF-D-10-05057.1.
[20] S. Migliorini, "Optimal ensemble-based selection of channels from advanced sounders in the presence of cloud," *Monthly Weather Rev.*, vol. 143, no. 9, pp. 3754–3773, Sep. 2015.
[21] F. I. Smith, "Improving the information content of IASI assimilation for numerical weather prediction," Ph.D. dissertation, Dept. Phys. Astron., Univ. Leicester, Leicester, U.K., 2015.
[22] C. E. Shannon and W. Weaver, "The mathematical theory of communication," Univ. Illinois Press, 1949.
[23] C. D. Rodgers, *Inverse Methods for Atmospheric Sounding: The Theory and Practice*. Singapore: World Scientific, 2000.
[24] D. Di, J. Li, W. Han, W. Bai, C. Wu, and W. P. Menzel, "Enhancing the fast radiative transfer model for FengYun-4 GIIRS by using local training profiles," *J. Geophys. Res., Atmos.*, vol. 123, no. 22, pp. 12–583, Nov. 2018.
[25] L. Garand *et al.*, "Radiance and Jacobian intercomparison of radiative transfer models applied to HIRS and AMSU channels," *J. Geophys. Res., Atmos.*, vol. 106, no. D20, pp. 24017–24031, Oct. 2001.
[26] J. R. Eyre, "A fast radiative transfer model for satellite sounding systems," Eur. Centre Medium Range Weather Forecasts, Reading, U.K., ECMWF Tech. Memorandum 176, 1991.
[27] R. Saunders, M. Matricardi, and P. Brunel, "An improved fast radiative transfer model for assimilation of satellite radiance observations," *Quart. J. Roy. Meteorolog. Soc.*, vol. 125, no. 556, pp. 1407–1425, Apr. 1999.
[28] R. Saunders *et al.*, "An update on the RTTOV fast radiative transfer model (currently at version 12)," *Geosci. Model Develop.*, vol. 11, no. 7, pp. 2717–2737, 2018, doi: 10.5194/gmd-11-2717-2018
[29] H. Hersbach and D. Dee, "ERA5 reanalysis is in production," *ECMWF Newslett.*, vol. 147, no. 7, p. 640, 2016.
[30] J. Li, W. W. Wolf, W. P. Menzel, W. Zhang, H.-L. Huang, and T. H. Achtor, "Global soundings of the atmosphere from ATOVS measurements: The algorithm and validation," *J. Appl. Meteorol.*, vol. 39, no. 8, pp. 1248–1268, Aug. 2000.
[31] S. A. Clough *et al.*, "Atmospheric radiative transfer modeling: A summary of the AER codes," *J. Quant. Spectrosc. Radiat. Transf.*, vol. 91, no. 2, pp. 233–244, Mar. 2005.
[32] Z. Li *et al.*, "Improving the understanding of CrIS full spectral resolution nonlocal thermodynamic equilibrium radiances using spectral correlation," *J. Geophys. Res., Atmos.*, vol. 125, no. 16, 2020, Art. no. e2020JD032710, doi: 2020JD032710.2020.
[33] Y. Xue, J. Li, Z. Li, M. M. Gunshor, and T. J. Schmit, "Evaluation of the diurnal variation of upper tropospheric humidity in reanalysis using homogenized observed radiances from international geostationary weather satellites," *Remote Sens.*, vol. 12, no. 10, p. 1628, May 2020.
[34] E. E. Borbas, S. W. Seemann, H. L. Huang, J. Li, and W. P. Menzel, "Global profile training database for satellite regression retrievals with estimates of skin temperature and emissivity," in *Proc. 14th Int. ATOVS Study Conf.* Beijing, China: CIMSS, 2005, pp. 763–770.
[35] A. Rodriguez, R. Stuhlmann, S. Tjemkes, D. M. Aminou, and P. Blythe, "Meteosat third generation: Mission and system concepts," *Proc. SPIE*, vol. 7453, no. 93, Sep. 2009, Art. no. 74530C.



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